

**Makespan**

**Lex: (Skills, Makespan)**

**Cognitive Load**

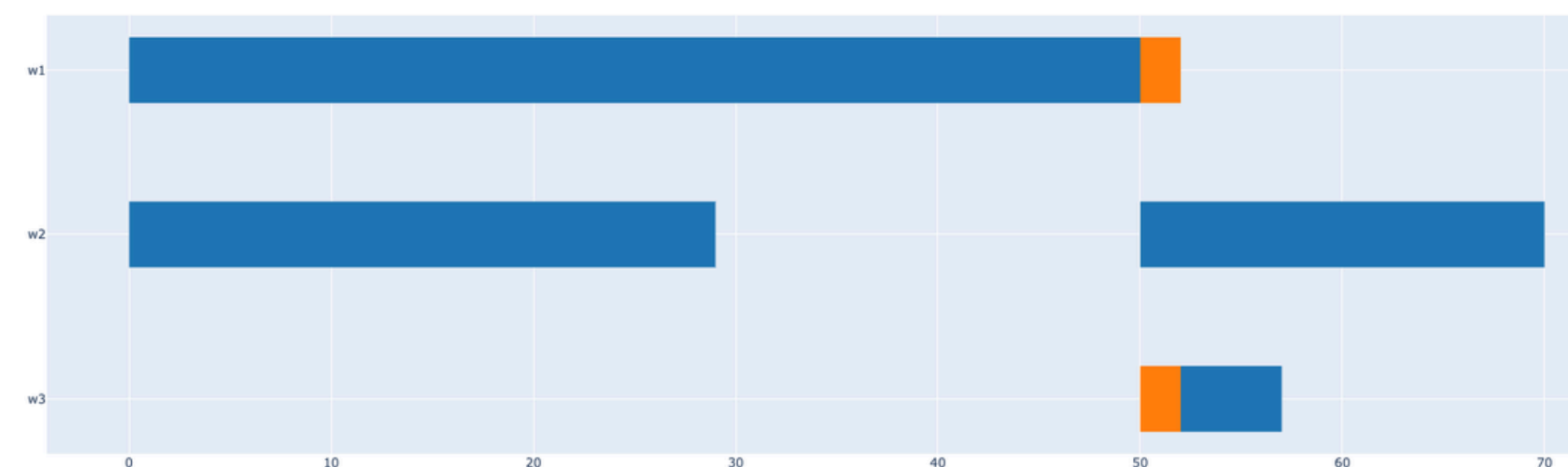
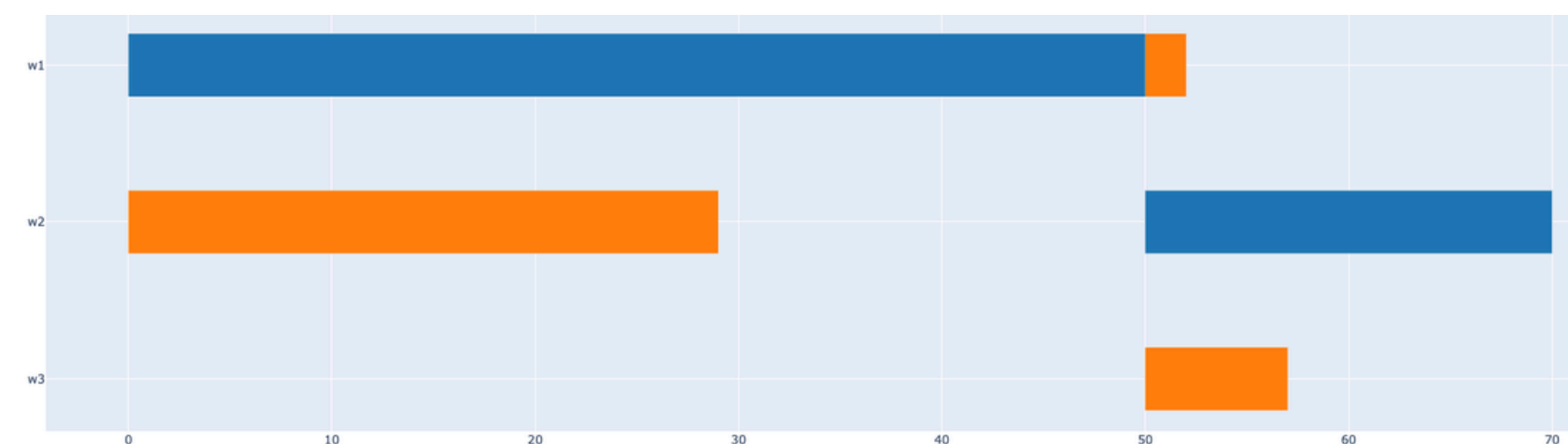
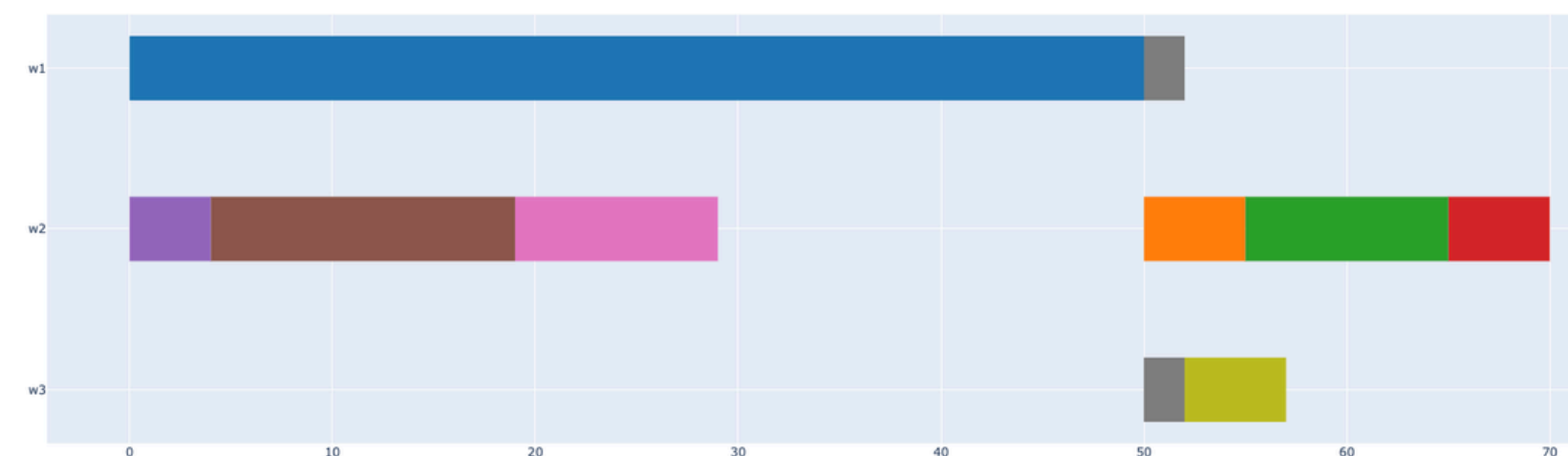
**Lex: (Makespan, Skills)**

```
Number of jobs: 2
Number of professions: 3
Number of tasks per profession: [2. 4. 1.]
Max number of operations per Jobs: 5
Total number of tasks: 7
Total number of workers: 3
Task to profession mapping:
{0: 0, 1: 0, 2: 1, 3: 1, 4: 1, 5: 1, 6: 2}
Worker levels: shape=(3, 3)
[[2. 2. 1.]
 [4. 3. 2.]
 [1. 2. 1.]]
Job difficulties: shape= (2,)
[2. 2.]
Task difficulties: shape= (7,)
[1 2 3 3 2 2 2]
Task processing times: shape= (7, 3)
[[ 50. 100. -1.]
 [ 15.  21. -1.]
 [  5.   8.  2.]
 [  4.   6.  2.]
 [  5.  10.  2.]
 [  4.  11.  2.]
 [ 10.  20.  2.]]
```

```
Jobs structure: len= 2
Job 0 :      0_(0,0) = 0      0_(0,1) = 2      0_(0,2) = 6      0_(0,3) = 2
Job 1 :      0_(1,0) = 3      0_(1,1) = 1      0_(1,2) = 6      0_(1,3) = 4      0_(1,4) = 4
```

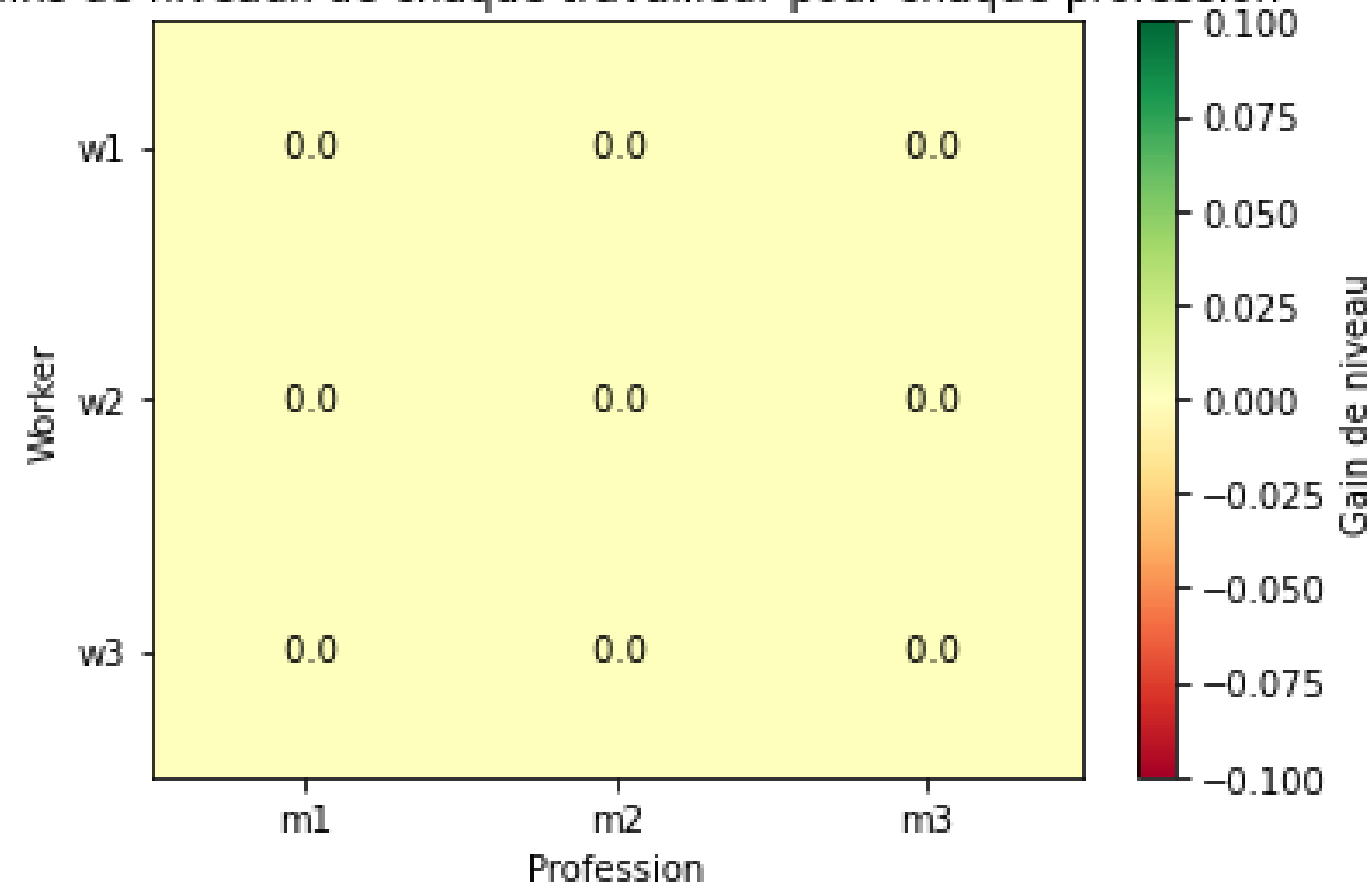
PS : oubli d'activer qu'un worker doit finir le job commencé :(

# Makespan



|   | Task | Start | Finish | Finish (var. f) | Processing time | Operation | Job | mode            | Level_w | Difficulty_task | Elementary task | Meter |
|---|------|-------|--------|-----------------|-----------------|-----------|-----|-----------------|---------|-----------------|-----------------|-------|
| 0 | w1   | 0.0   | 50.0   | 50.0            | 50.0            | (1,1)     | J1  | alone           | 2.0     | 1               | 0.0             | 0.0   |
| 1 | w1   | 50.0  | 52.0   | 52.0            | 2.0             | (2,4)     | J2  | collaboratively | 2.0     | 2               | 4.0             | 1.0   |
| 2 | w2   | 0.0   | 4.0    | 4.0             | 4.0             | (2,1)     | J2  | alone           | 3.0     | 3               | 3.0             | 1.0   |
| 3 | w2   | 4.0   | 19.0   | 19.0            | 15.0            | (2,2)     | J2  | alone           | 4.0     | 2               | 1.0             | 0.0   |
| 4 | w2   | 19.0  | 29.0   | 29.0            | 10.0            | (2,3)     | J2  | alone           | 2.0     | 2               | 6.0             | 2.0   |
| 5 | w2   | 50.0  | 55.0   | 55.0            | 5.0             | (1,2)     | J1  | alone           | 3.0     | 3               | 2.0             | 1.0   |
| 6 | w2   | 55.0  | 65.0   | 65.0            | 10.0            | (1,3)     | J1  | alone           | 2.0     | 2               | 6.0             | 2.0   |
| 7 | w2   | 65.0  | 70.0   | 70.0            | 5.0             | (1,4)     | J1  | alone           | 3.0     | 3               | 2.0             | 1.0   |
| 8 | w3   | 50.0  | 52.0   | 52.0            | 2.0             | (2,4)     | J2  | collaboratively | 2.0     | 2               | 4.0             | 1.0   |
| 9 | w3   | 52.0  | 57.0   | 57.0            | 5.0             | (2,5)     | J2  | alone           | 2.0     | 2               | 4.0             | 1.0   |

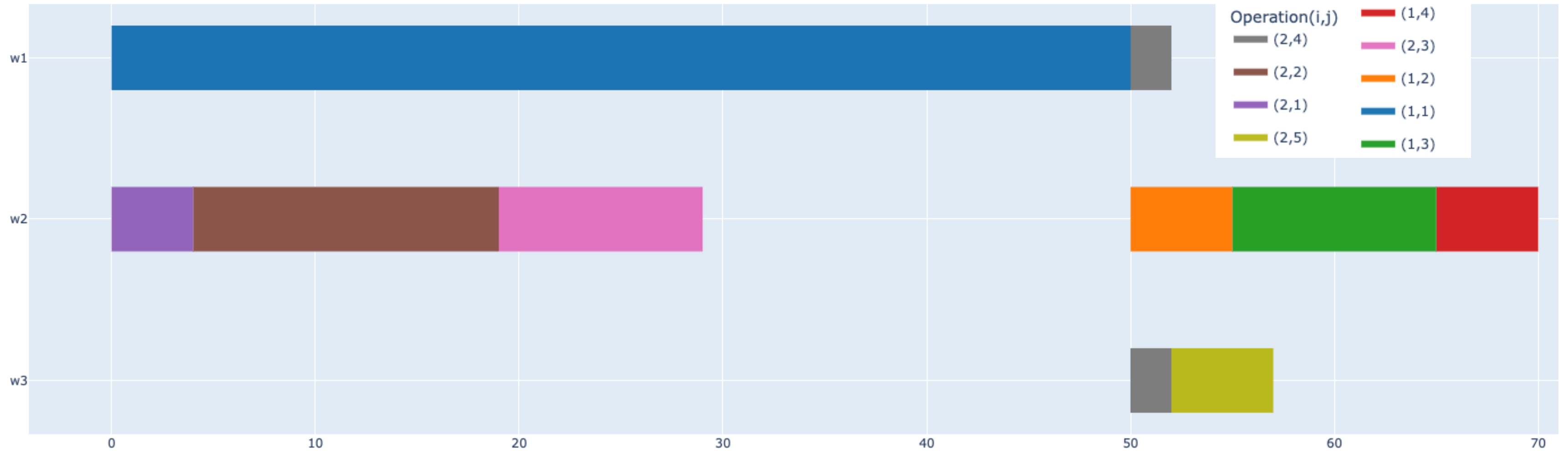
Gains de niveaux de chaque travailleur pour chaque profession



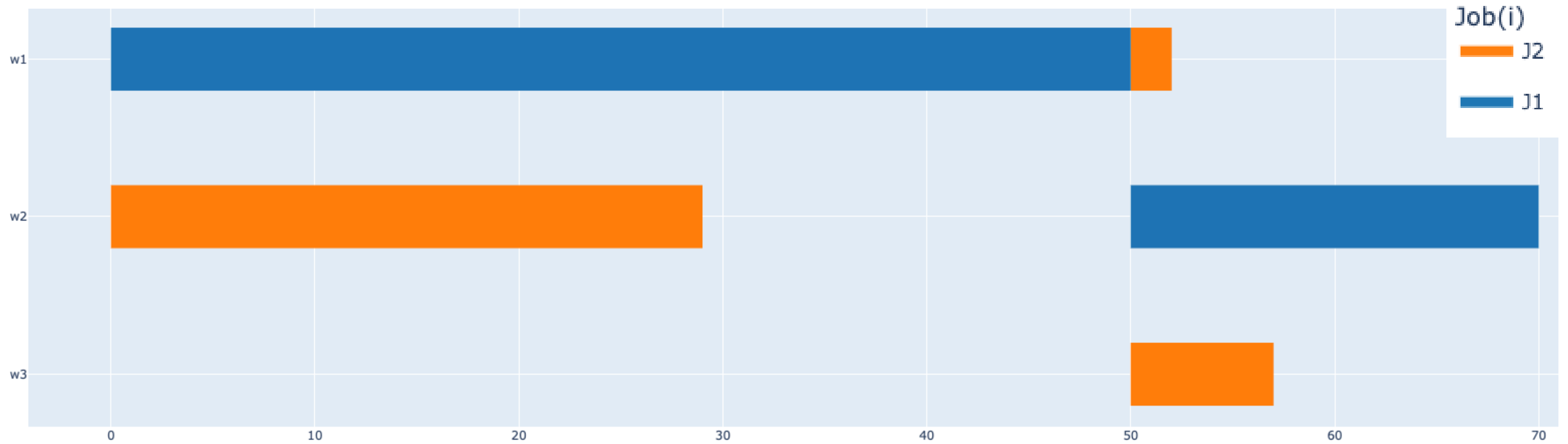
Aucun worker n'augmente ses skills

C\_max = 70

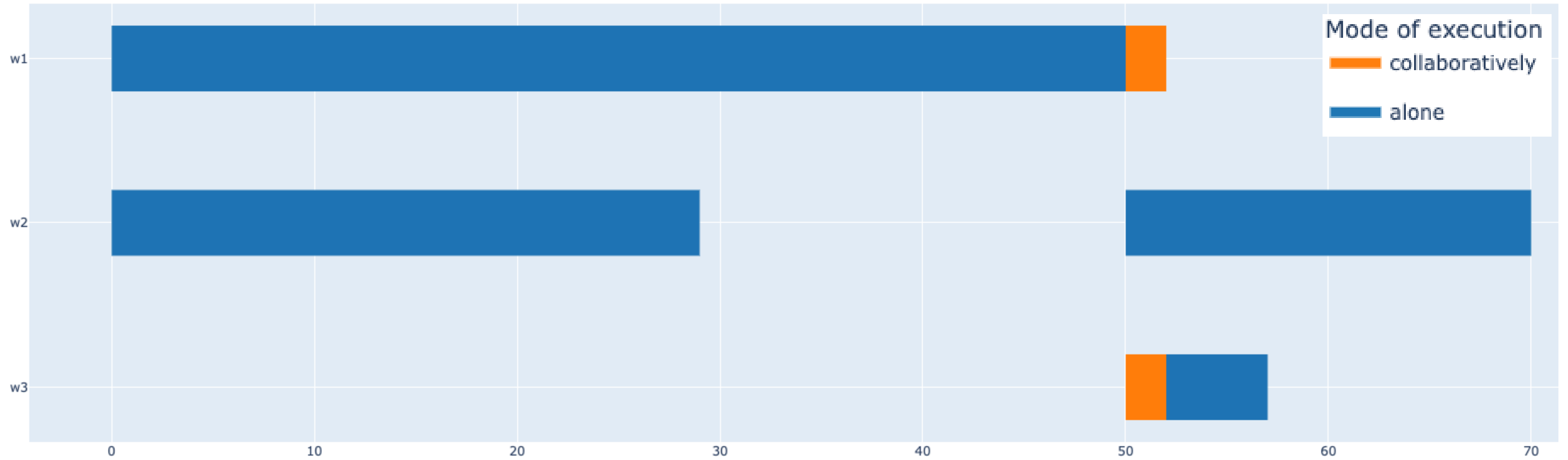
# Makespan



# Makespan



# Makespan



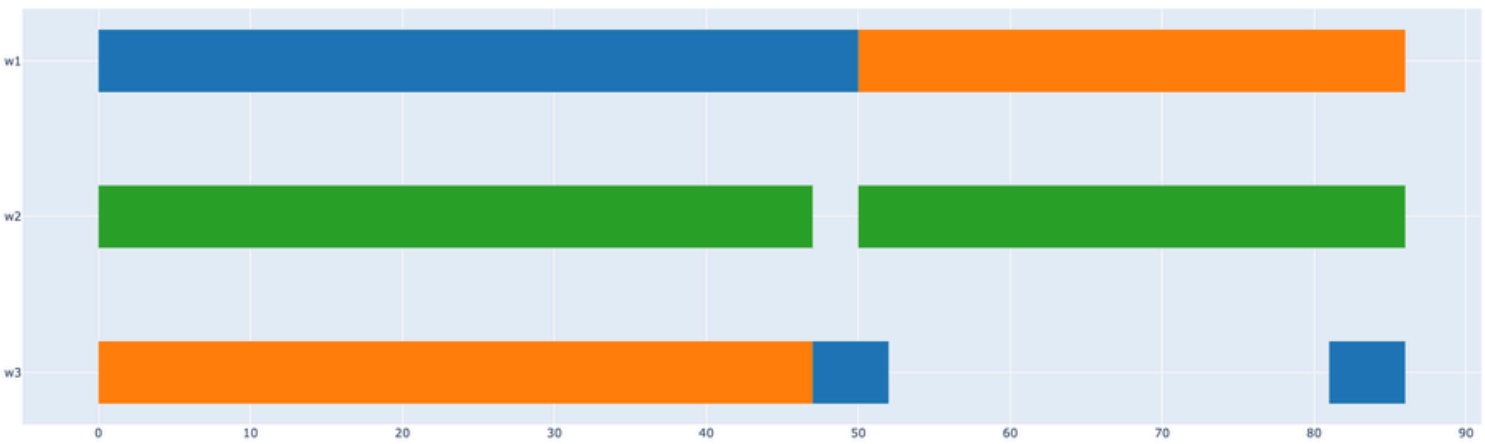
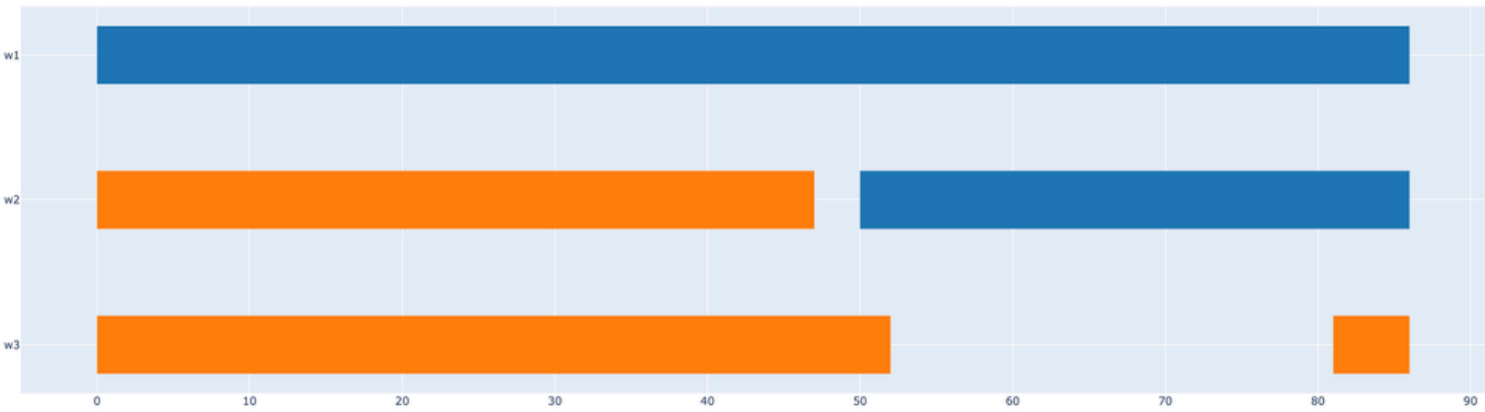
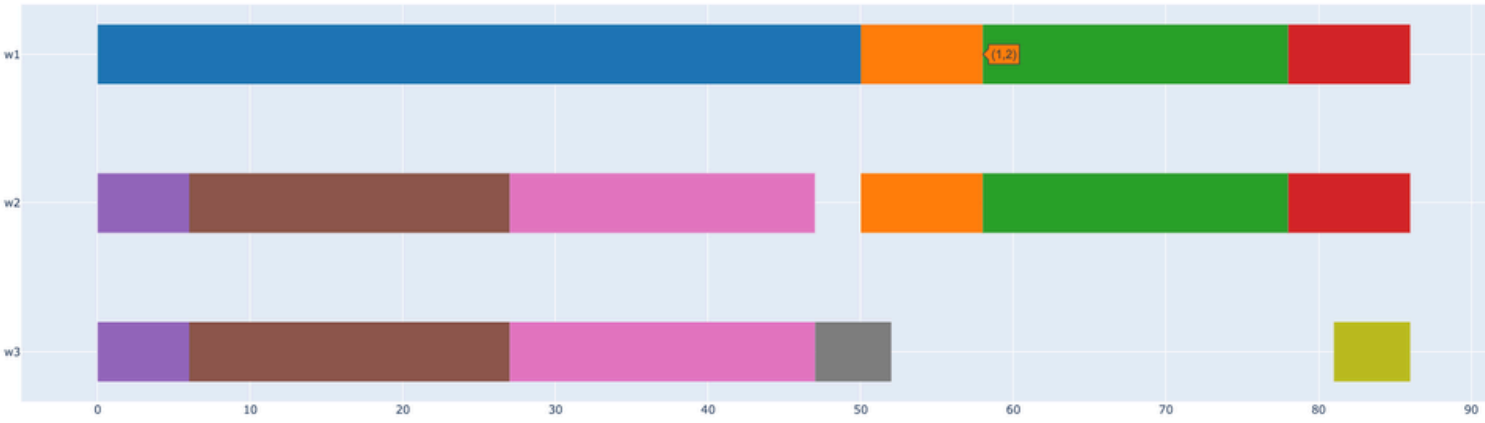


# Makespan

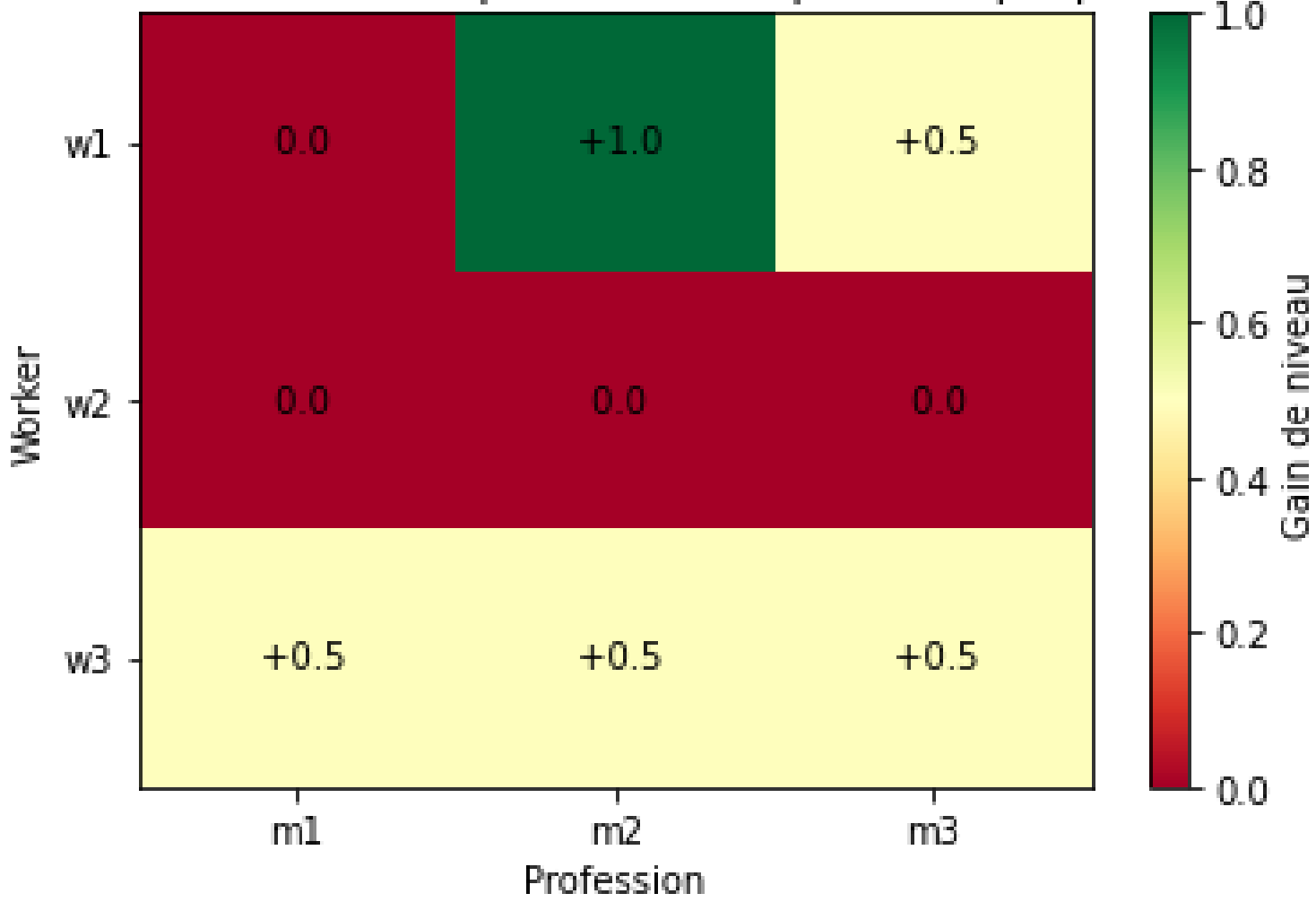
|   | Task | Start | Finish | Finish (var. f) | Processing time | Operation | Job | mode            | Level_w | Difficulty_task | Elementary task | Metier |
|---|------|-------|--------|-----------------|-----------------|-----------|-----|-----------------|---------|-----------------|-----------------|--------|
| 0 | w1   | 0.0   | 50.0   | 50.0            | 50.0            | (1,1)     | J1  | alone           | 2.0     | 1               | 0.0             | 0.0    |
| 1 | w1   | 50.0  | 52.0   | 52.0            | 2.0             | (2,4)     | J2  | collaboratively | 2.0     | 2               | 4.0             | 1.0    |
| 2 | w2   | 0.0   | 4.0    | 4.0             | 4.0             | (2,1)     | J2  | alone           | 3.0     | 3               | 3.0             | 1.0    |
| 3 | w2   | 4.0   | 19.0   | 19.0            | 15.0            | (2,2)     | J2  | alone           | 4.0     | 2               | 1.0             | 0.0    |
| 4 | w2   | 19.0  | 29.0   | 29.0            | 10.0            | (2,3)     | J2  | alone           | 2.0     | 2               | 6.0             | 2.0    |
| 5 | w2   | 50.0  | 55.0   | 55.0            | 5.0             | (1,2)     | J1  | alone           | 3.0     | 3               | 2.0             | 1.0    |
| 6 | w2   | 55.0  | 65.0   | 65.0            | 10.0            | (1,3)     | J1  | alone           | 2.0     | 2               | 6.0             | 2.0    |
| 7 | w2   | 65.0  | 70.0   | 70.0            | 5.0             | (1,4)     | J1  | alone           | 3.0     | 3               | 2.0             | 1.0    |
| 8 | w3   | 50.0  | 52.0   | 52.0            | 2.0             | (2,4)     | J2  | collaboratively | 2.0     | 2               | 4.0             | 1.0    |
| 9 | w3   | 52.0  | 57.0   | 57.0            | 5.0             | (2,5)     | J2  | alone           | 2.0     | 2               | 4.0             | 1.0    |

# Lex: (Skills, Makespan)

|    | Task | Start | Finish | Finish (var. f) | Processing time | Operation | Job | mode                  | Level_w | Difficulty_task | Elementary task | Metier |
|----|------|-------|--------|-----------------|-----------------|-----------|-----|-----------------------|---------|-----------------|-----------------|--------|
| 0  | w1   | 0.0   | 50.0   | 50.0            | 50.0            | (1,1)     | J1  | alone                 | 2.0     | 1               | 0.0             | 0.0    |
| 1  | w1   | 50.0  | 58.0   | 58.0            | 8.0             | (1,2)     | J1  | learning (apprentice) | 2.0     | 3               | 2.0             | 1.0    |
| 2  | w1   | 58.0  | 78.0   | 78.0            | 20.0            | (1,3)     | J1  | learning (apprentice) | 1.0     | 2               | 6.0             | 2.0    |
| 3  | w1   | 78.0  | 86.0   | 86.0            | 8.0             | (1,4)     | J1  | learning (apprentice) | 2.0     | 3               | 2.0             | 1.0    |
| 4  | w2   | 0.0   | 6.0    | 6.0             | 6.0             | (2,1)     | J2  | learning (tutor)      | 3.0     | 3               | 3.0             | 1.0    |
| 5  | w2   | 6.0   | 27.0   | 27.0            | 21.0            | (2,2)     | J2  | learning (tutor)      | 4.0     | 2               | 1.0             | 0.0    |
| 6  | w2   | 27.0  | 47.0   | 47.0            | 20.0            | (2,3)     | J2  | learning (tutor)      | 2.0     | 2               | 6.0             | 2.0    |
| 7  | w2   | 50.0  | 58.0   | 58.0            | 8.0             | (1,2)     | J1  | learning (tutor)      | 3.0     | 3               | 2.0             | 1.0    |
| 8  | w2   | 58.0  | 78.0   | 78.0            | 20.0            | (1,3)     | J1  | learning (tutor)      | 2.0     | 2               | 6.0             | 2.0    |
| 9  | w2   | 78.0  | 86.0   | 86.0            | 8.0             | (1,4)     | J1  | learning (tutor)      | 3.0     | 3               | 2.0             | 1.0    |
| 10 | w3   | 0.0   | 6.0    | 6.0             | 6.0             | (2,1)     | J2  | learning (apprentice) | 2.0     | 3               | 3.0             | 1.0    |
| 11 | w3   | 6.0   | 27.0   | 27.0            | 21.0            | (2,2)     | J2  | learning (apprentice) | 1.0     | 2               | 1.0             | 0.0    |
| 12 | w3   | 27.0  | 47.0   | 47.0            | 20.0            | (2,3)     | J2  | learning (apprentice) | 1.0     | 2               | 6.0             | 2.0    |
| 13 | w3   | 47.0  | 52.0   | 52.0            | 5.0             | (2,4)     | J2  | alone                 | 2.0     | 2               | 4.0             | 1.0    |
| 14 | w3   | 81.0  | 86.0   | 86.0            | 5.0             | (2,5)     | J2  | alone                 | 2.0     | 2               | 4.0             | 1.0    |



Gains de niveaux de chaque travailleur pour chaque profession

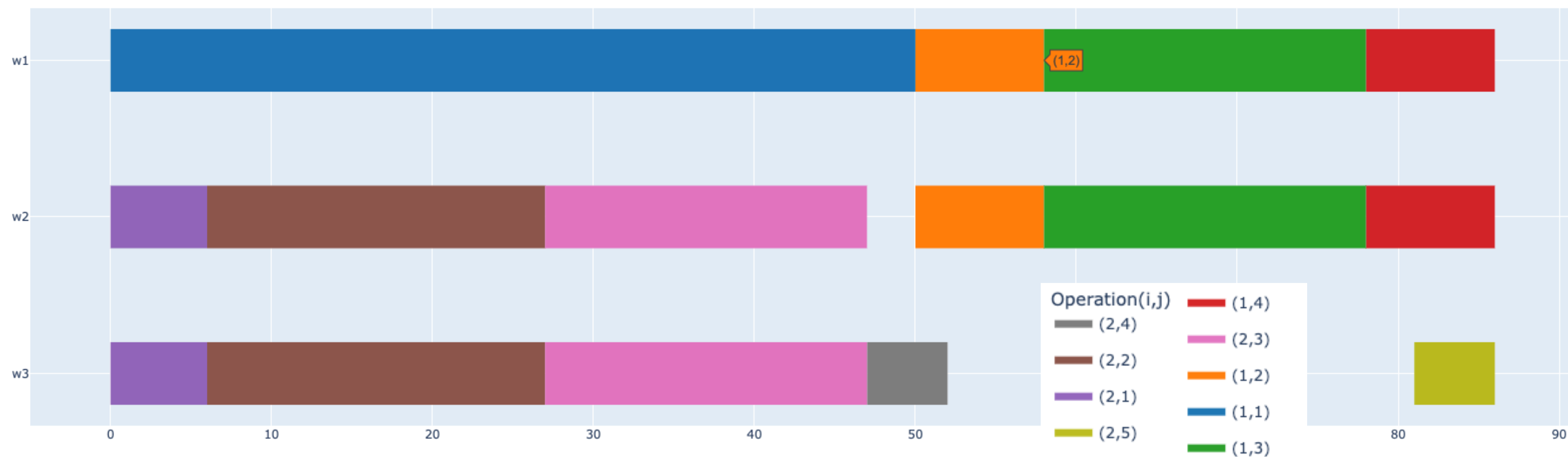


Des opérateurs ont augmenter leurs skills

- voir question d'équité car l'augmentation n'est pas forcément équilibrer (pour le moment somme des skills)

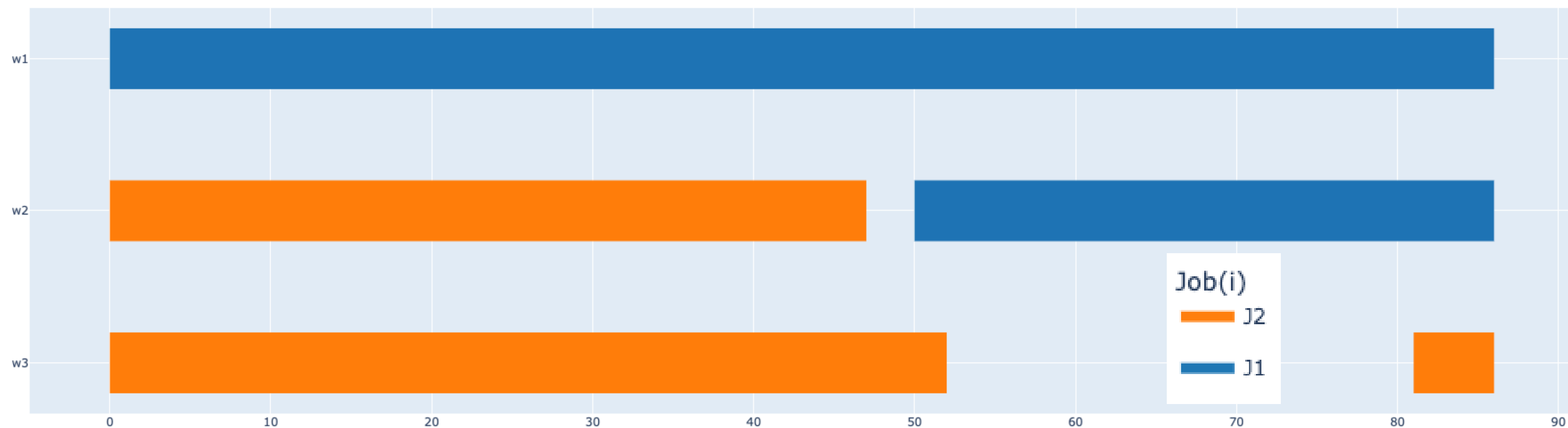
**C\_max = 86**

# Lex: (Skills, Makespan)

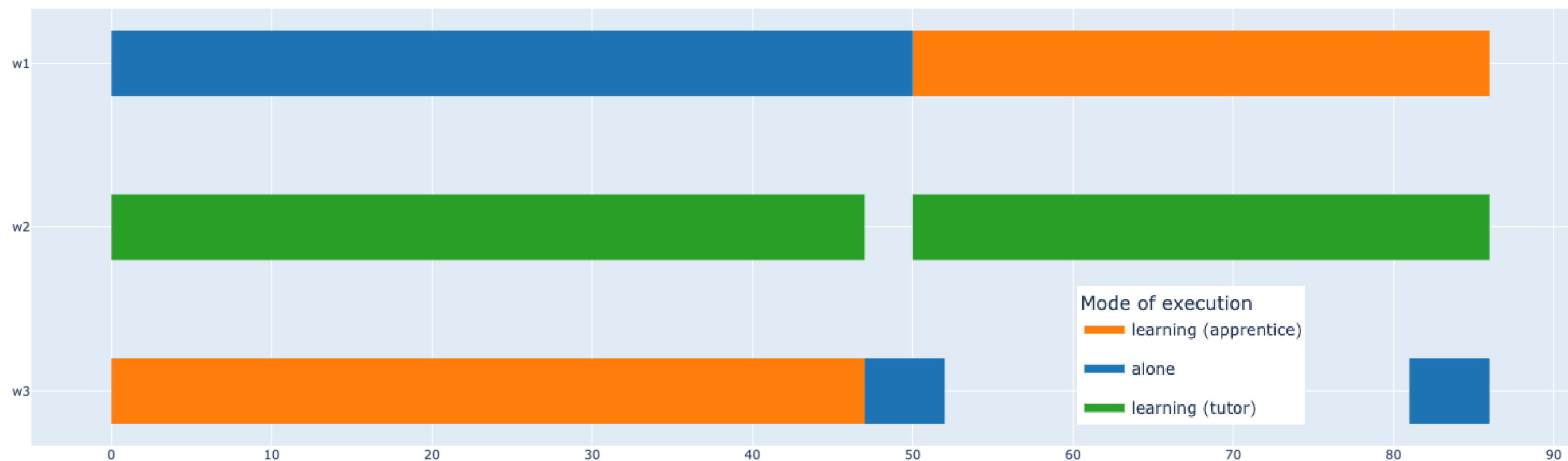




# Lex: (Skills, Makespan)



# Lex: (Skills, Makespan)

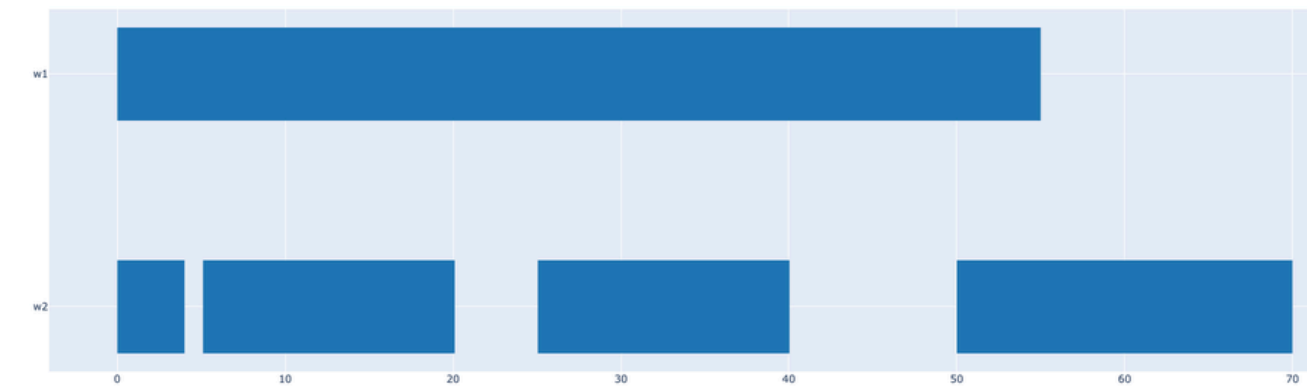
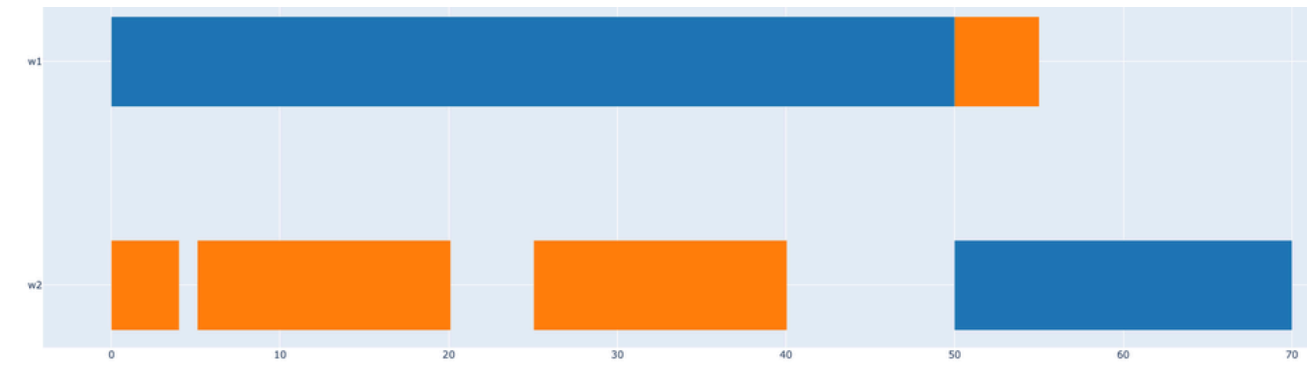
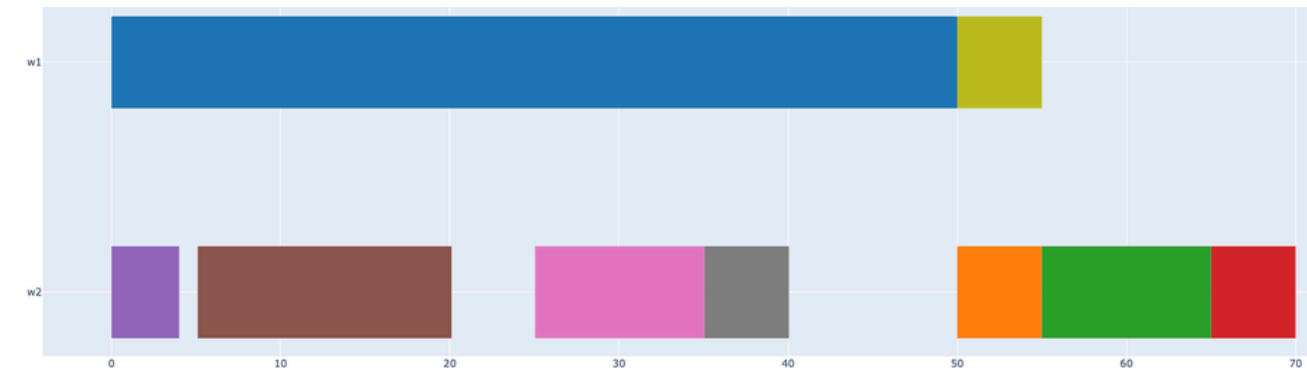


# Lex: (Skills, Makespan)

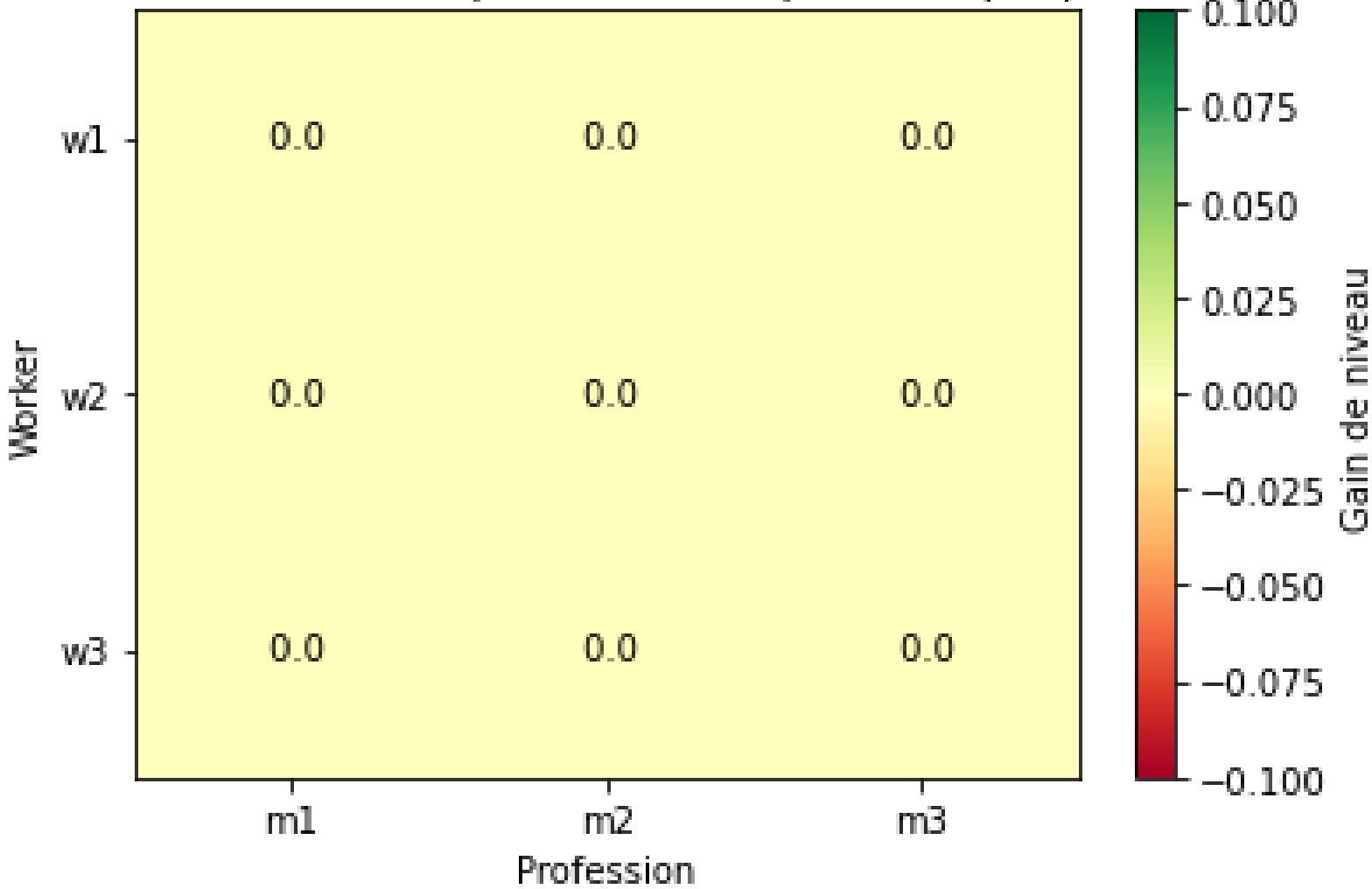
|    | Task | Start | Finish | Finish (var. f) | Processing time | Operation | Job | mode                  | Level_w | Difficulty_task | Elementary task | Metier |
|----|------|-------|--------|-----------------|-----------------|-----------|-----|-----------------------|---------|-----------------|-----------------|--------|
| 0  | w1   | 0.0   | 50.0   | 50.0            | 50.0            | (1,1)     | J1  | alone                 | 2.0     | 1               | 0.0             | 0.0    |
| 1  | w1   | 50.0  | 58.0   | 58.0            | 8.0             | (1,2)     | J1  | learning (apprentice) | 2.0     | 3               | 2.0             | 1.0    |
| 2  | w1   | 58.0  | 78.0   | 78.0            | 20.0            | (1,3)     | J1  | learning (apprentice) | 1.0     | 2               | 6.0             | 2.0    |
| 3  | w1   | 78.0  | 86.0   | 86.0            | 8.0             | (1,4)     | J1  | learning (apprentice) | 2.0     | 3               | 2.0             | 1.0    |
| 4  | w2   | 0.0   | 6.0    | 6.0             | 6.0             | (2,1)     | J2  | learning (tutor)      | 3.0     | 3               | 3.0             | 1.0    |
| 5  | w2   | 6.0   | 27.0   | 27.0            | 21.0            | (2,2)     | J2  | learning (tutor)      | 4.0     | 2               | 1.0             | 0.0    |
| 6  | w2   | 27.0  | 47.0   | 47.0            | 20.0            | (2,3)     | J2  | learning (tutor)      | 2.0     | 2               | 6.0             | 2.0    |
| 7  | w2   | 50.0  | 58.0   | 58.0            | 8.0             | (1,2)     | J1  | learning (tutor)      | 3.0     | 3               | 2.0             | 1.0    |
| 8  | w2   | 58.0  | 78.0   | 78.0            | 20.0            | (1,3)     | J1  | learning (tutor)      | 2.0     | 2               | 6.0             | 2.0    |
| 9  | w2   | 78.0  | 86.0   | 86.0            | 8.0             | (1,4)     | J1  | learning (tutor)      | 3.0     | 3               | 2.0             | 1.0    |
| 10 | w3   | 0.0   | 6.0    | 6.0             | 6.0             | (2,1)     | J2  | learning (apprentice) | 2.0     | 3               | 3.0             | 1.0    |
| 11 | w3   | 6.0   | 27.0   | 27.0            | 21.0            | (2,2)     | J2  | learning (apprentice) | 1.0     | 2               | 1.0             | 0.0    |
| 12 | w3   | 27.0  | 47.0   | 47.0            | 20.0            | (2,3)     | J2  | learning (apprentice) | 1.0     | 2               | 6.0             | 2.0    |
| 13 | w3   | 47.0  | 52.0   | 52.0            | 5.0             | (2,4)     | J2  | alone                 | 2.0     | 2               | 4.0             | 1.0    |
| 14 | w3   | 81.0  | 86.0   | 86.0            | 5.0             | (2,5)     | J2  | alone                 | 2.0     | 2               | 4.0             | 1.0    |

# Cognitive Load

|   | Task | Start     | Finish    | Finish (var. f) | Processing time | Operation | Job | mode  | Level_w | Difficulty_task | Elementary task | Metier |
|---|------|-----------|-----------|-----------------|-----------------|-----------|-----|-------|---------|-----------------|-----------------|--------|
| 0 | w1   | 0.000000  | 50.000000 | 50.000000       | 50.0            | (1,1)     | J1  | alone | 2.0     | 1               | 0.0             | 0.0    |
| 1 | w1   | 50.000000 | 55.000000 | 55.000000       | 5.0             | (2,5)     | J2  | alone | 2.0     | 2               | 4.0             | 1.0    |
| 2 | w2   | 0.000000  | 4.000000  | 4.000000        | 4.0             | (2,1)     | J2  | alone | 3.0     | 3               | 3.0             | 1.0    |
| 3 | w2   | 5.104239  | 20.104239 | 20.104239       | 15.0            | (2,2)     | J2  | alone | 4.0     | 2               | 1.0             | 0.0    |
| 4 | w2   | 25.044034 | 35.044034 | 35.044034       | 10.0            | (2,3)     | J2  | alone | 2.0     | 2               | 6.0             | 2.0    |
| 5 | w2   | 35.044034 | 40.044034 | 40.044034       | 5.0             | (2,4)     | J2  | alone | 3.0     | 2               | 4.0             | 1.0    |
| 6 | w2   | 50.000000 | 55.000000 | 55.000000       | 5.0             | (1,2)     | J1  | alone | 3.0     | 3               | 2.0             | 1.0    |
| 7 | w2   | 55.000000 | 65.000000 | 65.000000       | 10.0            | (1,3)     | J1  | alone | 2.0     | 2               | 6.0             | 2.0    |
| 8 | w2   | 65.000000 | 70.000000 | 70.000000       | 5.0             | (1,4)     | J1  | alone | 3.0     | 3               | 2.0             | 1.0    |



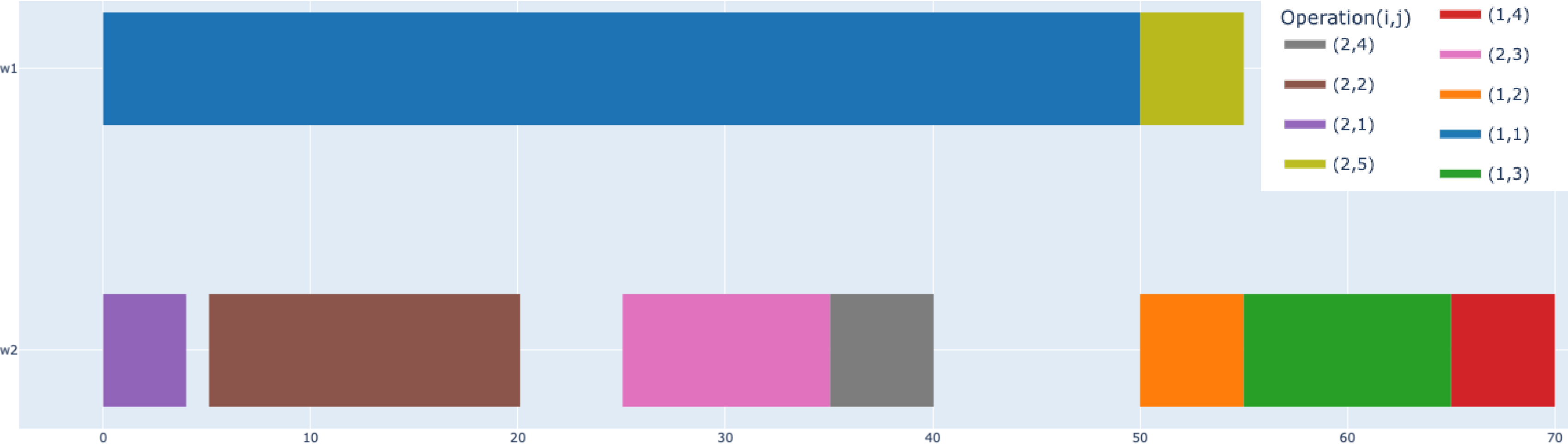
Gains de niveaux de chaque travailleur pour chaque profession



Aucun worker n'augmente ses skills

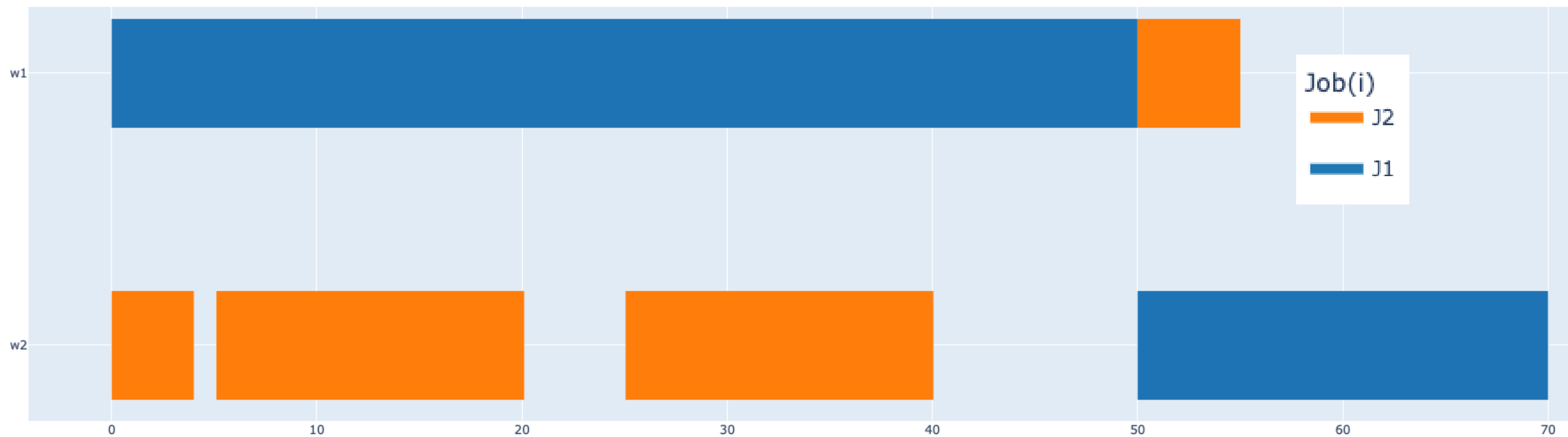
C\_max = 70

# Cognitive Load

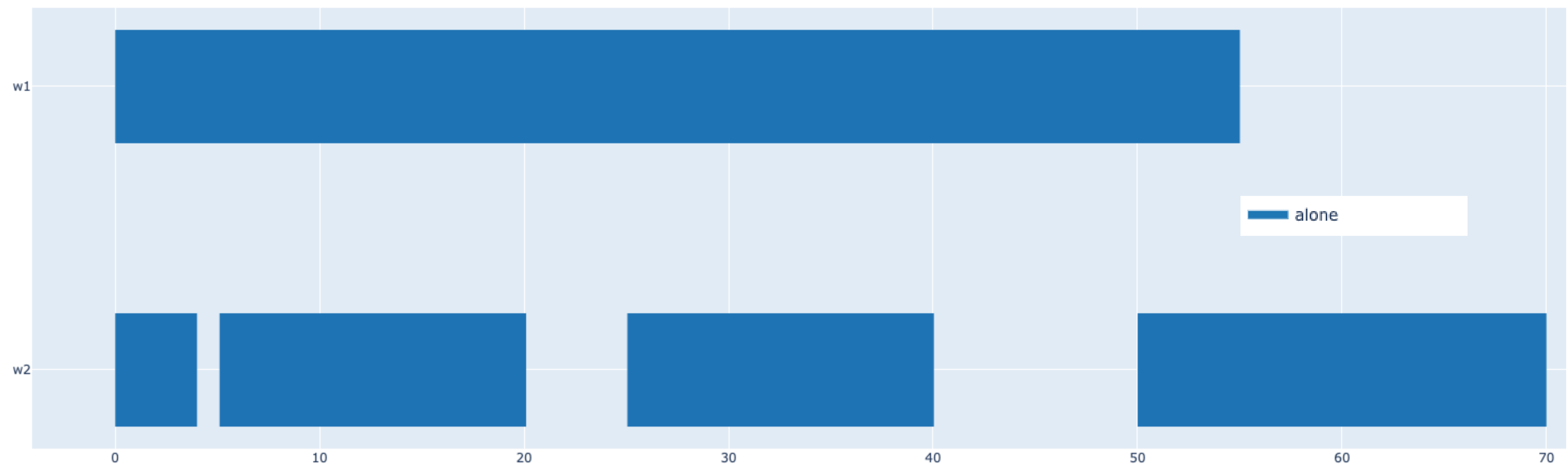




# Cognitive Load



# Cognitive Load

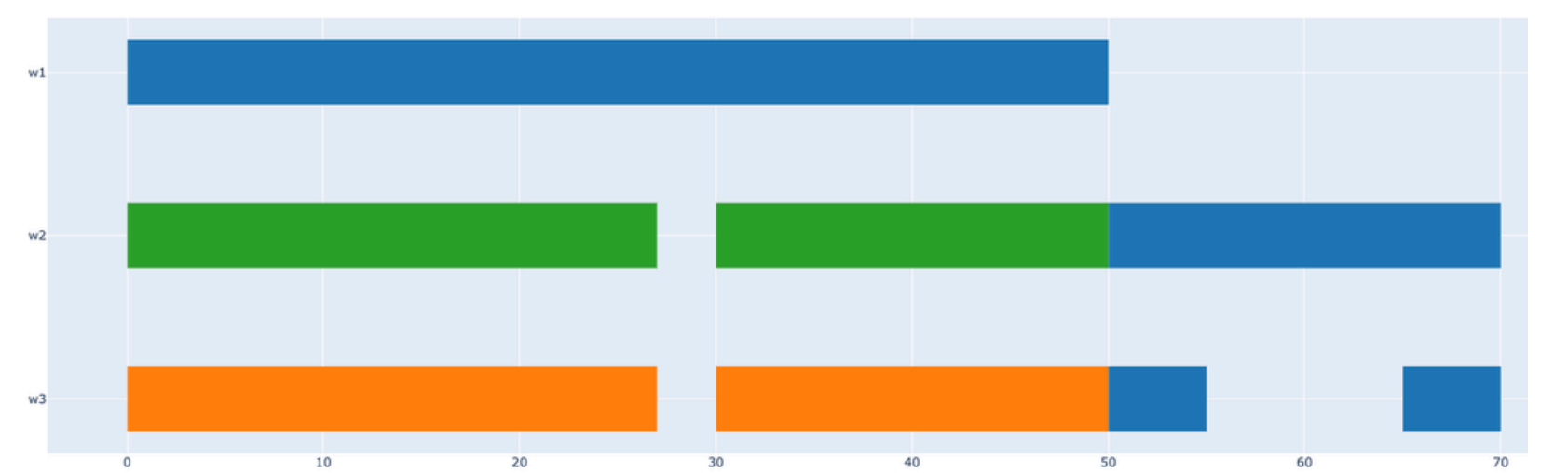
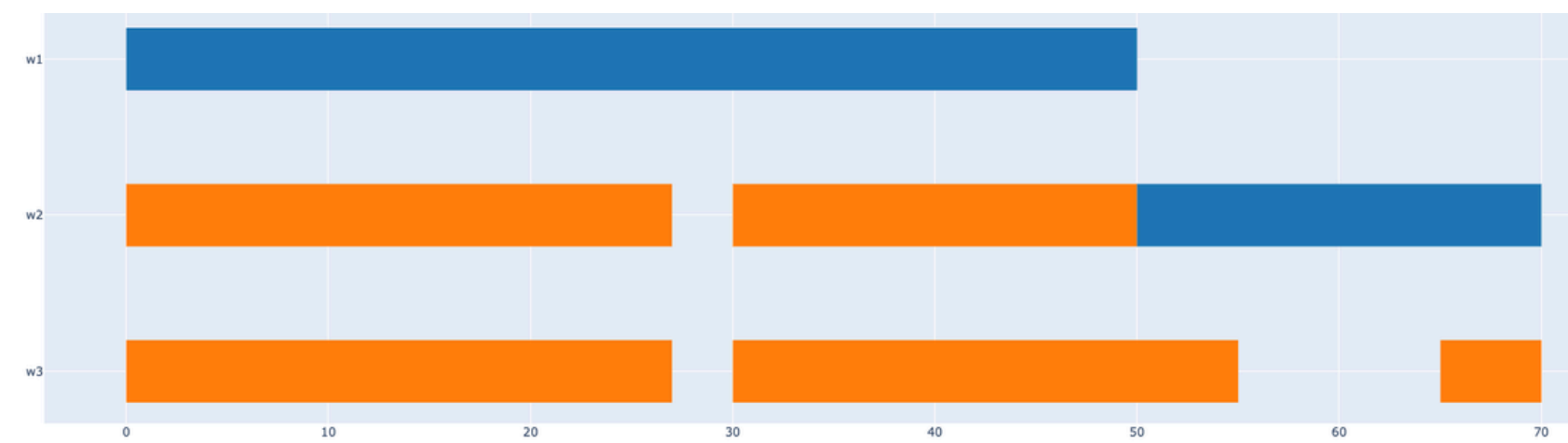
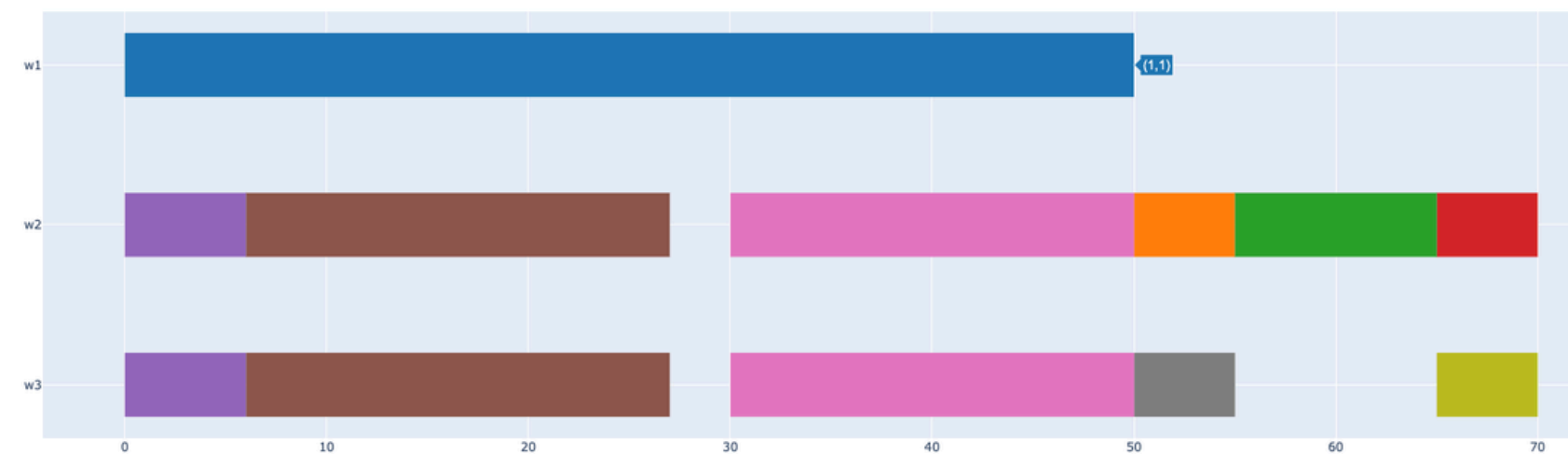


# Cognitive Load

|   | Task | Start     | Finish    | Finish (var. f) | Processing time | Operation | Job | mode  | Level_w | Difficulty_task | Elementary task | Metier |
|---|------|-----------|-----------|-----------------|-----------------|-----------|-----|-------|---------|-----------------|-----------------|--------|
| 0 | w1   | 0.000000  | 50.000000 | 50.000000       | 50.0            | (1,1)     | J1  | alone | 2.0     | 1               | 0.0             | 0.0    |
| 1 | w1   | 50.000000 | 55.000000 | 55.000000       | 5.0             | (2,5)     | J2  | alone | 2.0     | 2               | 4.0             | 1.0    |
| 2 | w2   | 0.000000  | 4.000000  | 4.000000        | 4.0             | (2,1)     | J2  | alone | 3.0     | 3               | 3.0             | 1.0    |
| 3 | w2   | 5.104239  | 20.104239 | 20.104239       | 15.0            | (2,2)     | J2  | alone | 4.0     | 2               | 1.0             | 0.0    |
| 4 | w2   | 25.044034 | 35.044034 | 35.044034       | 10.0            | (2,3)     | J2  | alone | 2.0     | 2               | 6.0             | 2.0    |
| 5 | w2   | 35.044034 | 40.044034 | 40.044034       | 5.0             | (2,4)     | J2  | alone | 3.0     | 2               | 4.0             | 1.0    |
| 6 | w2   | 50.000000 | 55.000000 | 55.000000       | 5.0             | (1,2)     | J1  | alone | 3.0     | 3               | 2.0             | 1.0    |
| 7 | w2   | 55.000000 | 65.000000 | 65.000000       | 10.0            | (1,3)     | J1  | alone | 2.0     | 2               | 6.0             | 2.0    |
| 8 | w2   | 65.000000 | 70.000000 | 70.000000       | 5.0             | (1,4)     | J1  | alone | 3.0     | 3               | 2.0             | 1.0    |

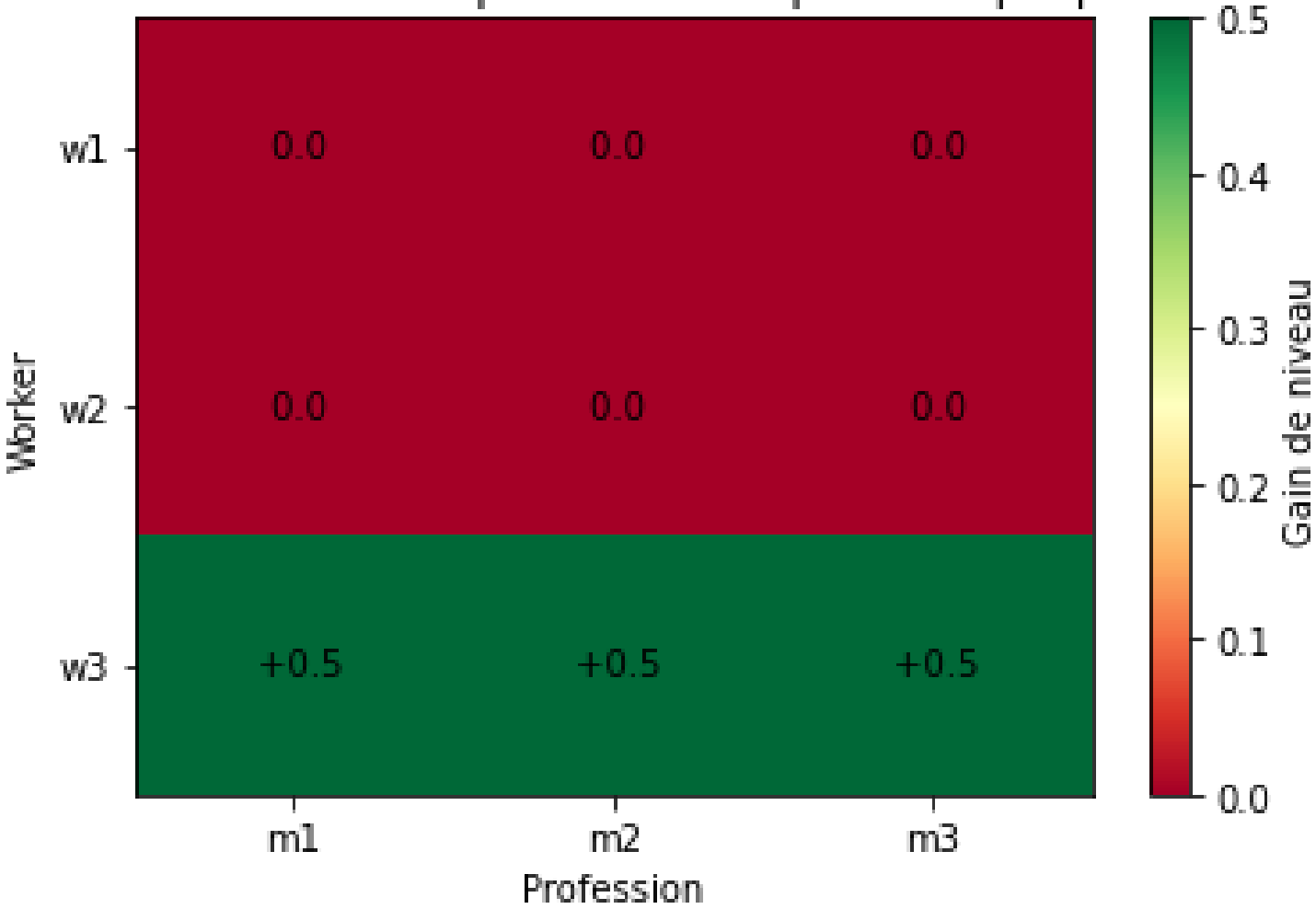


# Lex: (Makespan, Skills)



|    | Task | Start     | Finish    | Finish (var. f) | Processing time | Operation | Job | mode                  | Level_w | Difficulty_task | Elementary task | Metier |
|----|------|-----------|-----------|-----------------|-----------------|-----------|-----|-----------------------|---------|-----------------|-----------------|--------|
| 0  | w1   | 0.000000  | 50.000000 | 50.000000       | 50.0            | (1,1)     | J1  | alone                 | 2.0     | 1               | 0.0             | 0.0    |
| 1  | w2   | 0.000000  | 6.000000  | 6.000000        | 6.0             | (2,1)     | J2  | learning (tutor)      | 3.0     | 3               | 3.0             | 1.0    |
| 2  | w2   | 6.000000  | 27.000000 | 27.000000       | 21.0            | (2,2)     | J2  | learning (tutor)      | 4.0     | 2               | 1.0             | 0.0    |
| 3  | w2   | 30.007001 | 50.007001 | 50.007001       | 20.0            | (2,3)     | J2  | learning (tutor)      | 2.0     | 2               | 6.0             | 2.0    |
| 4  | w2   | 50.007001 | 55.007001 | 55.007001       | 5.0             | (1,2)     | J1  | alone                 | 3.0     | 3               | 2.0             | 1.0    |
| 5  | w2   | 55.007001 | 65.007001 | 65.007001       | 10.0            | (1,3)     | J1  | alone                 | 2.0     | 2               | 6.0             | 2.0    |
| 6  | w2   | 65.007001 | 70.007001 | 70.007001       | 5.0             | (1,4)     | J1  | alone                 | 3.0     | 3               | 2.0             | 1.0    |
| 7  | w3   | 0.000000  | 6.000000  | 6.000000        | 6.0             | (2,1)     | J2  | learning (apprentice) | 2.0     | 3               | 3.0             | 1.0    |
| 8  | w3   | 6.000000  | 27.000000 | 27.000000       | 21.0            | (2,2)     | J2  | learning (apprentice) | 1.0     | 2               | 1.0             | 0.0    |
| 9  | w3   | 30.007001 | 50.007001 | 50.007001       | 20.0            | (2,3)     | J2  | learning (apprentice) | 1.0     | 2               | 6.0             | 2.0    |
| 10 | w3   | 50.007001 | 55.007001 | 55.007001       | 5.0             | (2,4)     | J2  | alone                 | 2.0     | 2               | 4.0             | 1.0    |
| 11 | w3   | 65.007001 | 70.007001 | 70.007001       | 5.0             | (2,5)     | J2  | alone                 | 2.0     | 2               | 4.0             | 1.0    |

Gains de niveaux de chaque travailleur pour chaque profession

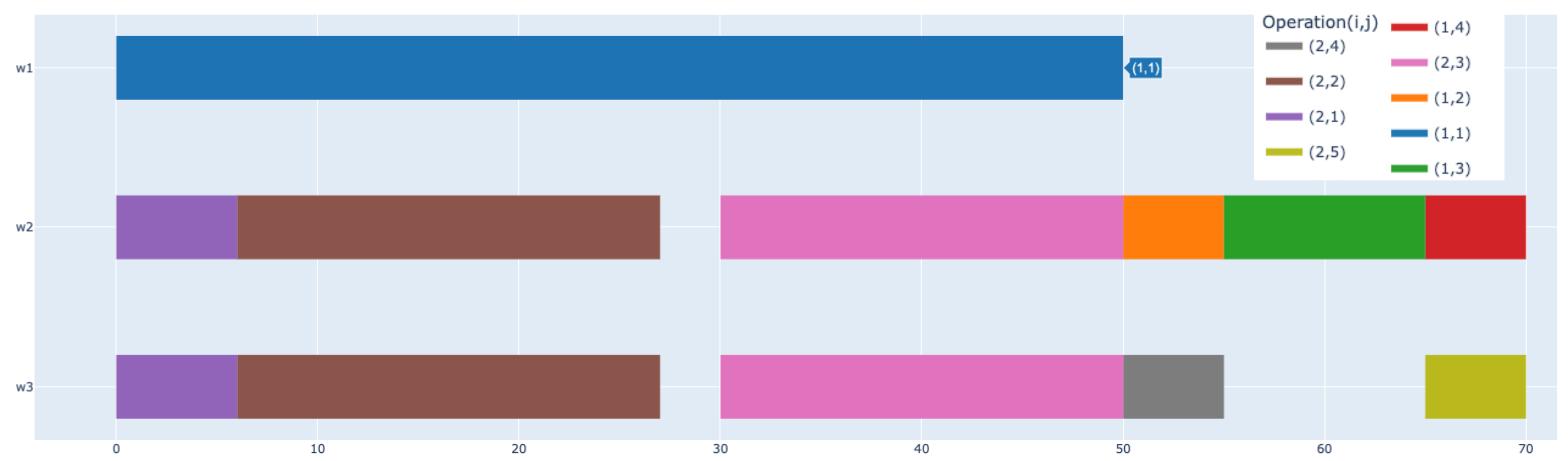


Des opérateurs ont augmenter leurs skills

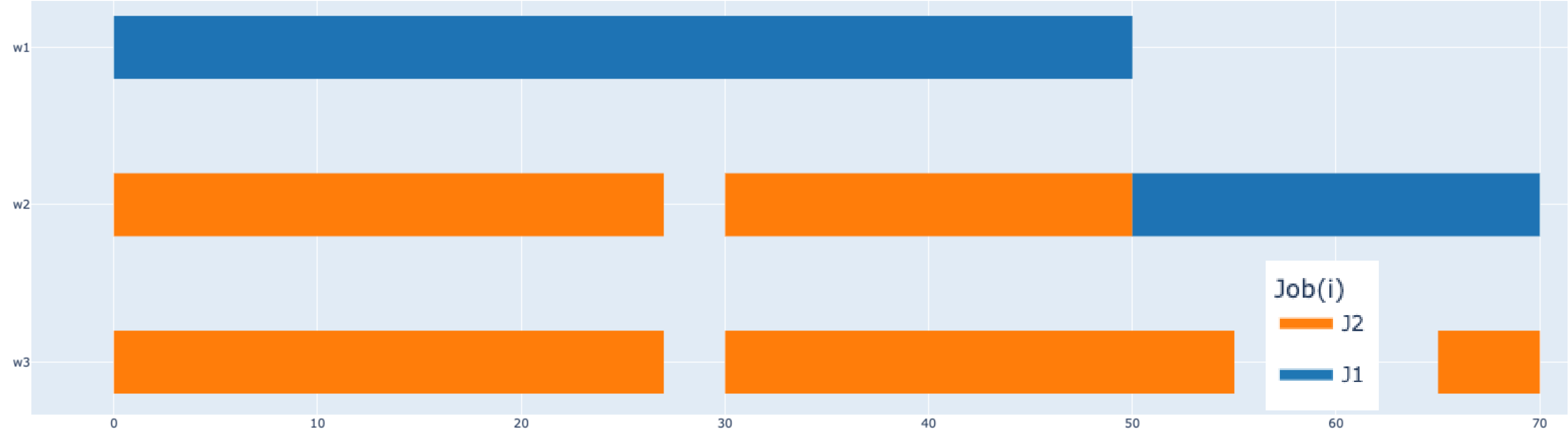
- voir question d'équité car l'augmentation n'est pas forcément équilibrer (pour le moment somme des skills)

C\_max = 70

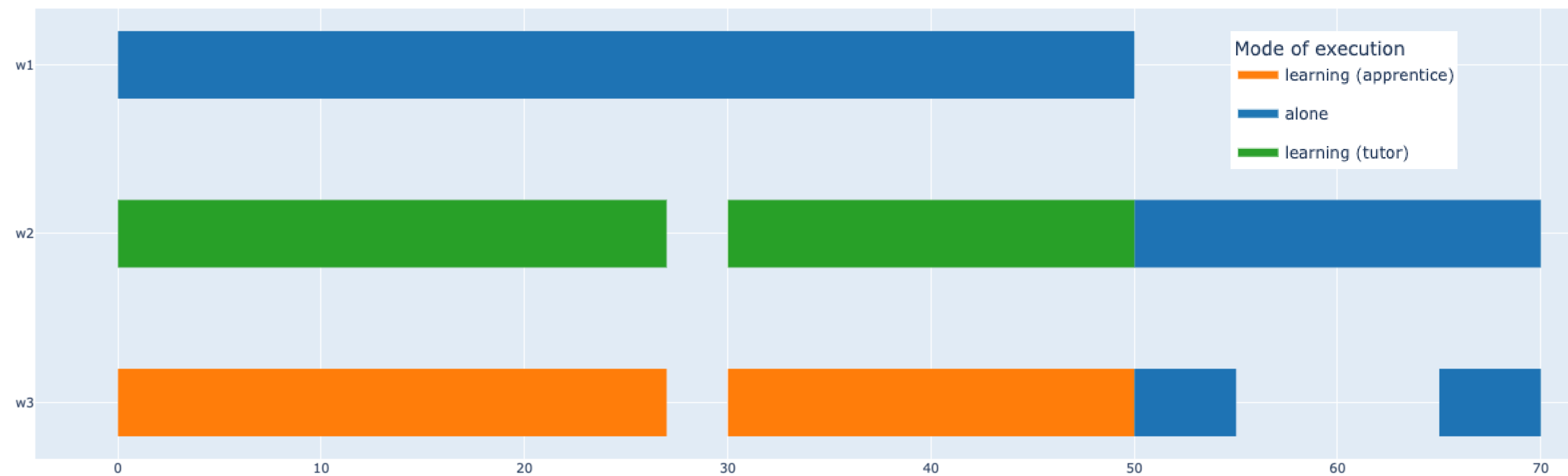
# Lex: (Makespan, Skills)



# Lex: (Makespan, Skills)



# Lex: (Makespan, Skills)



# Lex: (Makespan, Skills)

|    | Task | Start     | Finish    | Finish (var. f) | Processing time | Operation | Job | mode                  | Level_w | Difficulty_task | Elementary task | Metier |
|----|------|-----------|-----------|-----------------|-----------------|-----------|-----|-----------------------|---------|-----------------|-----------------|--------|
| 0  | w1   | 0.000000  | 50.000000 | 50.000000       | 50.0            | (1,1)     | J1  | alone                 | 2.0     | 1               | 0.0             | 0.0    |
| 1  | w2   | 0.000000  | 6.000000  | 6.000000        | 6.0             | (2,1)     | J2  | learning (tutor)      | 3.0     | 3               | 3.0             | 1.0    |
| 2  | w2   | 6.000000  | 27.000000 | 27.000000       | 21.0            | (2,2)     | J2  | learning (tutor)      | 4.0     | 2               | 1.0             | 0.0    |
| 3  | w2   | 30.007001 | 50.007001 | 50.007001       | 20.0            | (2,3)     | J2  | learning (tutor)      | 2.0     | 2               | 6.0             | 2.0    |
| 4  | w2   | 50.007001 | 55.007001 | 55.007001       | 5.0             | (1,2)     | J1  | alone                 | 3.0     | 3               | 2.0             | 1.0    |
| 5  | w2   | 55.007001 | 65.007001 | 65.007001       | 10.0            | (1,3)     | J1  | alone                 | 2.0     | 2               | 6.0             | 2.0    |
| 6  | w2   | 65.007001 | 70.007001 | 70.007001       | 5.0             | (1,4)     | J1  | alone                 | 3.0     | 3               | 2.0             | 1.0    |
| 7  | w3   | 0.000000  | 6.000000  | 6.000000        | 6.0             | (2,1)     | J2  | learning (apprentice) | 2.0     | 3               | 3.0             | 1.0    |
| 8  | w3   | 6.000000  | 27.000000 | 27.000000       | 21.0            | (2,2)     | J2  | learning (apprentice) | 1.0     | 2               | 1.0             | 0.0    |
| 9  | w3   | 30.007001 | 50.007001 | 50.007001       | 20.0            | (2,3)     | J2  | learning (apprentice) | 1.0     | 2               | 6.0             | 2.0    |
| 10 | w3   | 50.007001 | 55.007001 | 55.007001       | 5.0             | (2,4)     | J2  | alone                 | 2.0     | 2               | 4.0             | 1.0    |
| 11 | w3   | 65.007001 | 70.007001 | 70.007001       | 5.0             | (2,5)     | J2  | alone                 | 2.0     | 2               | 4.0             | 1.0    |





